Hypothesis Testing for One Sample Proportion

Statistics Worksheet · Grade 11–12 / Intro College Stats

Name: _____ Date: _____

Learning Objectives

- Write null and alternative hypotheses for a one-sample proportion test
- Verify conditions (normality, independence, randomness) before conducting a test
- Calculate the z-test statistic and use the p-value to draw a conclusion

Problems

1. A company claims that 60% of its customers are satisfied with their service. Identify the null and alternative hypotheses if a researcher wants to test whether the true proportion differs from 60%.

$$H_0 : p = 0.60 \quad H_a : p \neq 0.60$$

2. A politician claims that more than 50% of voters support her. Write the null and alternative hypotheses for this claim.

$$H_0 : p = 0.50 \quad H_a : p > 0.50$$

3. A sample of 80 students is drawn. The hypothesized population proportion is p = 0.40. Check whether the normality condition is satisfied using the rule: n·p > 10 and n·q > 10.

$$n = 80, \quad p = 0.40, \quad q = 1 - 0.40$$

4. A survey of 50 people finds that 18 prefer brand A. Calculate the sample proportion (p-hat).

$$\hat{p} = \frac{x}{n} = \frac{18}{50}$$

5. Using a sample of n = 100 with p-hat = 0.55 and a hypothesized population proportion of p = 0.50, calculate the z-test statistic.

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

6. Ryan, a basketball player, made 40% of his free throws last season. This season he made 25 out of 40 attempts. Test at the 1% significance level whether his proportion has improved. First, write the hypotheses and check the normality condition.

$$H_0 : p = 0.40 \quad H_a : p > 0.40$$

7. Continuing the Ryan example: p = 0.40, n = 40, and he made 25 out of 40 free throws. Calculate the z-test statistic.

$$z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}} = \frac{\frac{25}{40} - 0.40}{\sqrt{\frac{(0.40)(0.60)}{40}}}$$

8. Using the z-test statistic of approximately 3.23 from the Ryan free throw problem and a right-tailed test at the 1% significance level (alpha = 0.01, critical z = 2.326), state the conclusion. The p-value for z = 3.23 is approximately 0.0006.

$$\alpha = 0.01, \quad z = 3.23, \quad p\text{-value} \approx 0.0006$$

9. A health agency claims that less than 25% of adults in a city exercise regularly. A random sample of 200 adults finds that 44 exercise regularly. At the 5% significance level, test the agency's claim. Show all four steps: (1) hypotheses, (2) conditions, (3) test statistic, (4) conclusion. The p-value for this test is approximately 0.1335.

$$H_0 : p = 0.25 \quad H_a : p < 0.25$$

10. A manufacturer claims that exactly 90% of its light bulbs last more than 1,000 hours. A quality inspector randomly tests 150 bulbs and finds that 126 last more than 1,000 hours. At the 5% significance level, test whether the true proportion differs from 90%. The p-value for this test is approximately 0.0241. State all four steps, identify the type of test, and write a full conclusion in context.

$$H_0 : p = 0.90 \quad H_a : p \neq 0.90$$

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Hypothesis Testing for One Sample Proportion — Answer Key

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Answer Key

1. Answer: H_0: p = 0.60 (two-tailed test)

- The null hypothesis always uses an equals sign: H_0: p = 0.60
- Because the researcher tests whether the proportion 'differs' (not simply higher or lower), the alternative is two-tailed: H_a: p ≠ 0.60

2. Answer: H_0: p = 0.50, H_a: p > 0.50 (right-tailed test)

- The null hypothesis states the proportion equals the claimed value: H_0: p = 0.50
- Because the claim is 'more than,' the alternative hypothesis is right-tailed: H_a: p > 0.50

3. Answer: Normality condition is satisfied: np = 32 > 10 and nq = 48 > 10

- Compute q = 1 – 0.40 = 0.60
- n·p = 80 × 0.40 = 32 > 10 ✓
- n·q = 80 × 0.60 = 48 > 10 ✓
- Both conditions are met, so normality is satisfied.

4. Answer: p-hat = 0.36

- Divide the number of successes by the sample size: $\hat{p}$ = 18 ÷ 50
- $\hat{p}$ = 0.36

5. Answer: z ≈ 1.00

- Identify values: $\hat{p}$ = 0.55, p = 0.50, q = 0.50, n = 100
- Compute standard deviation: √(0.50 × 0.50 / 100) = √(0.0025) = 0.05
- z = (0.55 – 0.50) / 0.05 = 0.05 / 0.05 = 1.00

6. Answer: H_a: p > 0.40 (right-tailed); np = 16 > 10, nq = 24 > 10 — normality satisfied

- The parameter p = proportion of Ryan's successful free throws
- H_0: p = 0.40 (no improvement); H_a: p > 0.40 (improvement) — right-tailed
- Check normality: n·p = 40 × 0.40 = 16 > 10 ✓ and n·q = 40 × 0.60 = 24 > 10 ✓
- Normality condition is satisfied.

7. Answer: z ≈ 3.23

- $\hat{p}$ = 25/40 = 0.625
- Standard deviation = √(0.40 × 0.60 / 40) = √(0.24/40) = √0.006 ≈ 0.07746
- z = (0.625 – 0.40) / 0.07746 = 0.225 / 0.07746 ≈ 3.23

8. Answer: Reject H_0. There is sufficient evidence that Ryan's free throw proportion has improved.

- Compare the p-value to α: 0.0006 < 0.01

• Since p-value < α, we reject the null hypothesis.
• Conclusion: At the 1% significance level, there is sufficient evidence to conclude that Ryan's free throw proportion has increased above 40%.

9. Answer: Fail to reject H_0; insufficient evidence that fewer than 25% exercise regularly

• Step 1 — Hypotheses: H_0: p = 0.25, H_a: p < 0.25 (left-tailed); p = proportion of adults who exercise regularly
• Step 2 — Conditions: n·p = 200×0.25 = 50 > 10 ✓; n·q = 200×0.75 = 150 > 10 ✓; sample is random ✓
• Step 3 — Test statistic: $\hat{p}$ = 44/200 = 0.22; SE = $\sqrt{(0.25\times0.75/200)}$ = $\sqrt{(0.0009375)} \approx 0.03062$; z = (0.22–0.25)/0.03062 ≈ –0.98
• Step 4 — Conclusion: p-value ≈ 0.1335 > 0.05 = α; fail to reject H_0. There is not sufficient evidence to support the claim.

10. Answer: Reject H_0; sufficient evidence that the true proportion differs from 90%

• Step 1 — Hypotheses: H_0: p = 0.90, H_a: p ≠ 0.90 (two-tailed); p = proportion of bulbs lasting more than 1,000 hours
• Step 2 — Conditions: n·p = 150×0.90 = 135 > 10 ✓; n·q = 150×0.10 = 15 > 10 ✓; random sample ✓
• Step 3 — Test statistic: $\hat{p}$ = 126/150 = 0.84; SE = $\sqrt{(0.90\times0.10/150)}$ = $\sqrt{(0.0006)} \approx 0.02449$; z = (0.84–0.90)/0.02449 ≈ –2.45
• Step 4 — Conclusion: p-value ≈ 0.0241 < 0.05 = α; reject H_0. At the 5% significance level, there is sufficient evidence that the true proportion of bulbs lasting more than 1,000 hours differs from the manufacturer's claimed 90%.

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